A3-PS027-TC108 · AIORI Hackathon 3

IPv6 Website and Dependency Readiness Auditor

Team: First_Timers

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October 13, 2026

Problem Description

- A website may have IPv6 DNS support but still rely on IPv4-only resources.
- Scripts, images, stylesheets, APIs and third-party services can affect IPv6 readiness.
- DNS-only checks do not reveal whether actual page resources load successfully.
- A practical audit should identify failed dependencies and the stage of failure.

Background

- IPv4 and IPv6 are internet protocols used to connect to web servers.
- Modern pages request resources from many domains, not just the main website.
- A connection can fail during DNS resolution, TCP connection, TLS setup or HTTP exchange.
- Browser-based observation plus network measurements gives a fuller picture of readiness.

Literature & RFC Review

- RFC 8200: foundational IPv6 protocol specification.
- RFC 8305: Happy Eyeballs v2 explains dual-stack connection behavior.
- Playwright documentation: browser automation and request observation.
- MDN Web Docs: resource loading and network timing concepts.
- Our proposed focus: combine page dependency discovery with IPv4/IPv6 stage-wise tests.

Proposed Solution

- Input a target website URL and load the page in an automated browser.
- Discover requested scripts, images, stylesheets, APIs and third-party domains.
- Measure DNS availability and IPv4/IPv6 connection behavior for discovered hosts.
- Classify failures by stage and link them to affected webpage dependencies.
- Display readiness findings in a dashboard and generate a summary report.

Optimization Proposed

Planned improvements over a basic DNS-only check

Aspect	Basic check	Proposed auditor
Scope	Main domain DNS	Page and third-party dependencies
Testing	DNS record presence	DNS, TCP, TLS and HTTP stages
Output	Simple pass/fail	Failure categories and dependency graph
Presentation	Raw check result	Dashboard and readable report

Solution Architecture & Design

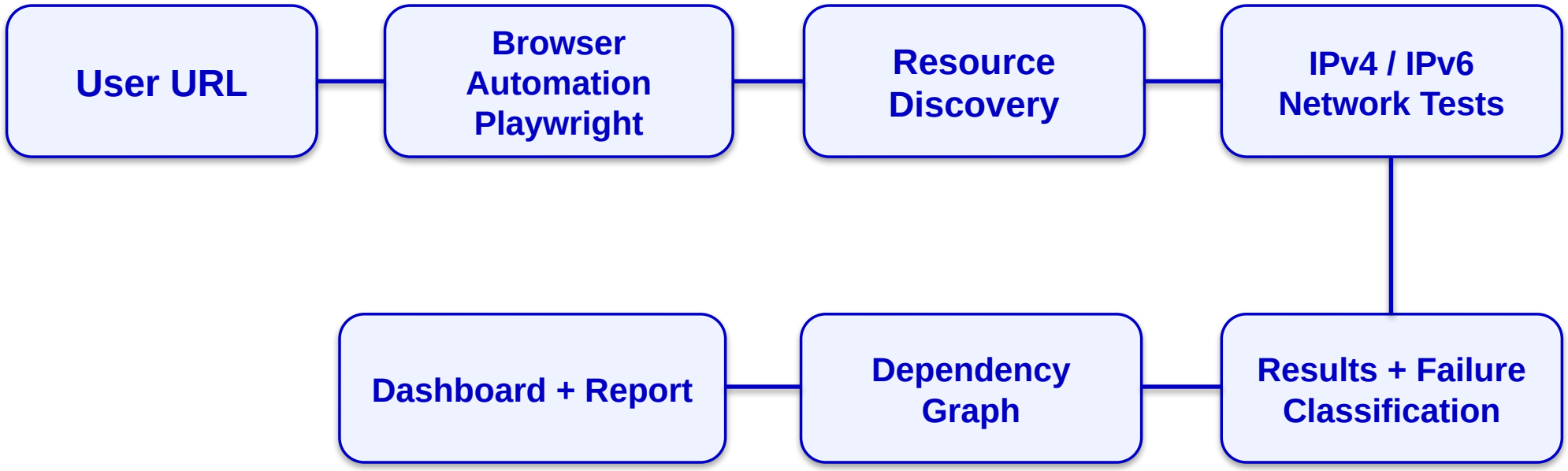


Figure: Planned high-level system architecture

Data Flow Diagram

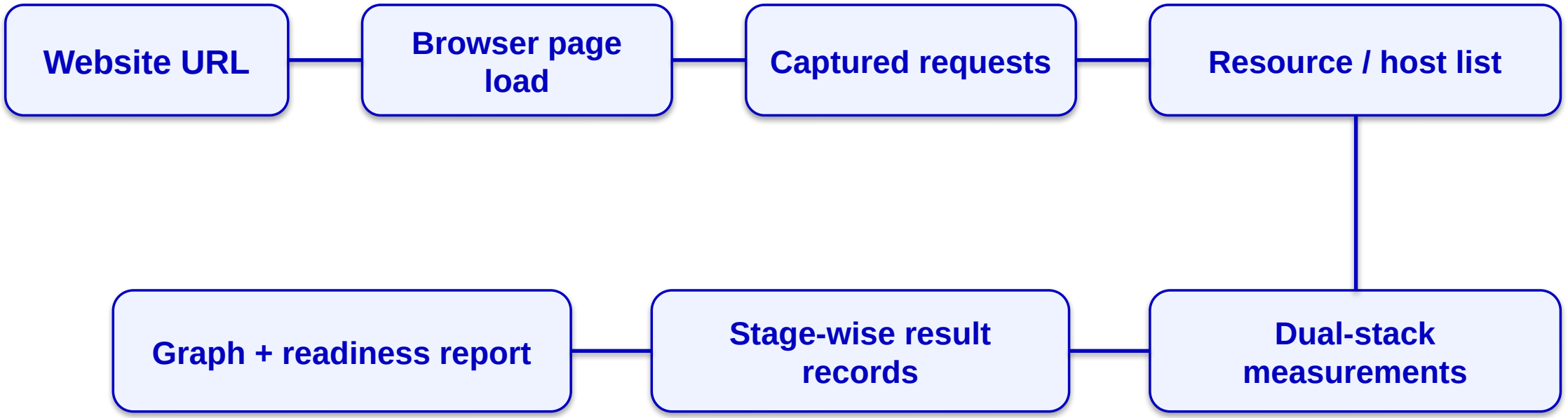


Figure: Planned data flow from webpage loading to report generation

Deployment Timeline & Team Plan

- Phase 1 — Understand requirements, define test cases and design the workflow.
- Phase 2 — Implement browser resource discovery and IPv4/IPv6 network checks.
- Phase 3 — Integrate dependency graph, result classification and dashboard.
- Phase 4 — Test on selected websites, fix issues and prepare a demonstration.

Proposed division of work (to be finalized by the team)

Nachan HaazalAli Danish — browser automation, resource discovery and integration.

Eshaan Brahmi — network measurements, result analysis and visualization.

Both members: GitHub collaboration, testing, documentation and presentation.

Bibliography

- IETF RFC 8200 — Internet Protocol, Version 6 (IPv6) Specification.
- IETF RFC 8305 — Happy Eyeballs Version 2: Better Connectivity Using Concurrency.
- Playwright documentation — https://playwright.dev/docs/intro
- MDN Web Docs — https://developer.mozilla.org/



Thank You for Your Attention!

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